

Tutorial - 1

In this tutorial, you will use one of the programming languages that you are familiar with such as C or FORTRAN and try to run the given codes for the following:

1. Consider a random matrix A of size $N \times N$ (user input), numerically show that $(A + A^T)$ is symmetric.
2. Adapt the function to *multiply two matrices* to perform a *matrix* $(N \times N)$ *vector* $(N \times 1)$ *product*. Plot the time taken as a function of N for $N = 64, 128, 256$ and 512 .

Linux basics

- `vi`, `vim`, `gedit`, `emacs`, `xemacs`, `vscode`, `gedit` – Editors to write and navigate code.
- `TeX` – To typeset documents.
- `Libre office` – Document writer, spread sheets and presentations.

Simple programs in C/C++/Fortran

1. Basics
 - Your first program
2. Array operations
 - Create and retrieve elements
 - Array addition
 - Dot product
3. Matrix operations
 - Create and retrieve elements
 - Matrix Addition
 - Matrix Multiplication

Plotting tool – gnuplot

- Line plot
 - `Document viewer` – viewer for images, pdf files etc.
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